JET TRANSFORMATION BLUEPRINT

Smart-Lean Bio-Thermal Infrastructure

Off-Grid Agricultural Industrial Asset Strategy

The Unified Operational Thesis

A high-yielding livestock environment cannot exist on standard grid-tied structures. Volatile highveld winter fronts and pathogenic vulnerabilities demand a fully closed-loop architecture.

Structural Defense

Isolating and grouping biological populations physically prevents systemic disease transitions.

Energy Sovereignty

Co-generating sub-surface thermal steam and BESS volts eliminates grid heating dependencies.

Physical Infrastructure Footprint

Infrastructure Asset	Building Sizing	Standard Capacity Target	Primary Environmental Shield
Poultry Sprint Barn	384 m^2 (12m × 32m)	5,760 Broilers / Batch	Ridge-Vent passive siphon chimney
Swine Stage 1 (Nursery)	48 m^2 (4m × 12m)	160 Piglets / Batch	Cavity insulated airtight walls
Swine Stage 2 (Finisher)	384 m^2 (12m × 32m)	160 Hogs / Batch	Open-sided curtain-assisted flow

30 kW / 30 kWh BESS Architecture

Integrated Energy Core

Designed to handle high-inductive starting currents from solar hydrology pumps and mechanical feed augers while maintaining safe reserves.

- **30 kW Inverter:** Manages peak daily motor and system startups safely.
- **30 kWh LFP Storage:** Provides deep-cycle nighttime lighting and automation reserve power.
- **Solar Array Interface:** High-yield monocrystalline roof panels drive continuous day charging.

Active Load Shield

Functions as the main electrical buffer, keeping the farm's automated controls and biological sensors active during grid blackout sequences.

Closed-Loop Biogas Thermal Engine

Volts & Steam Co-Generation

Our nighttime heating strategy does not stress the electrical batteries. We recycle biological manure into sovereign fuel:

- **Methane Recovery:** Anaerobic digestors convert slurry waste into clean biogas.
- **Electrical Feed:** Biogas GenSets feed volts directly to the BESS at night.
- **Sub-Floor Thermal Plumb:** Exhaust jacket heat routes hot steam beneath the nursery floors.

Zero-Electricity Heating

By using engine steam heat for floor warmth, the heavy heating baseload is removed from the battery banks entirely during cold winter drops.

Poultry Barn Passive Aerodynamics

Convective Ventilation Engine

Automated tunnel extraction fans draw heavy currents that threaten batteries. We design for zero-watt passive air circulation:

- **18-Degree Roof Pitch:** Generates a passive siphon chimney effect naturally.
- **Ridge-Vent Openings:** Hot, metabolic air rises and exits the top center continuously.
- **PIR Insulated Ceilings:** 50mm insulation blocks extreme sun heat from reaching broilers.

Convective Flow

The open-sided layout leverages natural cross-winds, ensuring steady oxygen exchange while preserving 100% of the BESS battery bank.

Storm & Rain Water Protection

Winch PVC Curtains

550g/m^2 heavy PVC curtains seal the sides within 90 seconds, stopping highveld rain spray instantly.

Dwarf Wall Block

A 600mm plastered masonry brick wall blocks surface water ingress, keeping nesting environments dry.

Overhanging Eaves

Generous roof margins divert heavy rain cascades far away from structural ventilation slots.

Poultry Brooding (Days 1 - 14)

Zonal Brooding Lockdown

Vulnerable day-old chicks require a strict 33°C. Heating the entire barn is highly inefficient, so we isolate the space:

- **Drop Curtains:** Thick internal PVC barriers isolate exactly one-third (128 m^2) of the building.
- **Volume Collapse:** Compressing the active airspace reduces the initial heating load by 66%.
- **Day 14 Release:** Curtains rise and birds expand into the full 384 m^2 footprint at 15 birds/m^2.

Step-Down Dynamics

As birds grow, target temperatures step down toward 24°C, relieving auxiliary heating demands from the microgrid.

Swine Pipeline Structure Sizing

Why do we build separate structures (4m × 12m Nursery vs 12m × 32m Finisher) instead of a combined block? We design for immune boundaries and thermodynamic isolation.

Nursery (Stage 1)

Demands draft-free 32°C. Elevated plastic slatted floors keep piglets hygienic during their critical immunity gap.

Finisher (Stage 2)

Demands cross-ventilation and 18°C. Heavy metabolic loads would cause heat stress in tight insulated structures.

The 4-Pod Immunological Firewall

Theory of Unwelcome

Housing nursery animals in one large open room allows viruses to sweep through the herd. Our standalone building acts as a firewall:

- **Partition Walls:** Solid 150mm masonry brick walls isolate four independent pods.
- **Vapor-Tight Doors:** Gasket-sealed insulated doors stop airborne pathogen transitions.
- **Capped Exposure:** If a virus enters Pod 1, the physical block isolates Pods 2, 3, and 4 completely.

25% Risk Cap

By structurally isolating the nursery block, pathogen breaches are capped at a maximum of 25%, saving 75% of your live capital asset.

Nursery Pod Microclimate Engineering

Sub-Floor Hygiene System

Cold concrete floor slabs conduct heat away from young piglets, causing hypothermia. We engineer for maximum body heat retention:

- **Polypropylene Slats:** Interlocking plastic flooring keeps piglets dry and thermally comfortable.
- **400mm Flush Gutter:** Animal waste drops down instantly through slats into a concrete flush channel.
- **Biogas Steam Tubes:** Steam lines deliver radiant warmth directly beneath the piglet resting zone.

Zonal Warmth

Each pod operates on its own steam valve. You only heat the specific rooms holding animals, keeping energy draws incredibly lean.

Throughput & Batch Velocity Math

480

Hogs Per Annum

The Staggered Throughput Flow

The compact 48 m^2 nursery block is optimized for rapid occupant rotation, avoiding underutilized space:

- **Staggered Waves:** 160 piglets per batch spent exactly 6 weeks in the nursery pods.
- **Active Transition:** Pigs march-over to the finisher barn, freeing up nursery space.
- **The Batch Equation:** 160 pigs × 3 cycles per year = 480 market-ready hogs.

The Hydrological Lifeline Sizing

Hydration & Slurry Moat

Livestock assets collapse instantly during dry spells. Our hydrological blueprint secures absolute operational continuity:

- **Solar Borehole Pumping:** Low-induction DC pumps draw exclusively on daytime solar energy.
- **Elevated Water Tower:** High tower layout distributes water naturally using gravity.
- **5-Day Safety Buffer:** Multi-tank storage reserves hold days of backup supply during outages.

Digester Moisture

Consistent, automated flow controls supply clean water lines to maintain moisture levels in active anaerobic digestion loops.

Water Sanitization Shield

Purification Barrier	Technology Deployed	Target Vectors Defeated	BESS Energy Impact
Mechanical Stage	50-Micron Sediment Disc	Silt, sand, and particulate scale	Zero-Watt Mechanical Flow
Biological Stage	Inline Ultraviolet (UV) Tube	Bacterial colonies & waterborne viral strains	Low-Voltage DC Current
Chemical Residual	Kinetic Chlorine Dosing	Biofilm and pathogen build-up in pipelines	Zero-Watt Proportional Flow

Gravity Slurry & Sump Topography

Pump-Free Waste Logistics

Mechanical pumps handling raw hog manure draw high starting currents and suffer from frequent blockages. We use physical gravity:

- **2% Floor Gradient:** All under-slat concrete channels are poured with a strict 1-in-50 downward drop.
- **Hydrostatic Flow:** Liquid manure fractions move naturally toward low sumps.
- **Sump Positioning:** Main sumps sit at the lowest physical site elevation to feed digesters.

Slurry Head Pressure

Leverages hydraulic head pressure to push slurry directly into digestion tanks without drawing a single watt of battery power.

Biosecurity Perimeter Barriers

Vehicle Spray Boot

Automatic spray nozzles at the main gate disinfect delivery truck wheels, neutralizing tracking vectors instantly.

Shower-In, Shower-Out

Forces personnel to shower and change into sanitized, farm-allocated uniforms before entering clean sections.

Localized Footbaths

Disinfecting boot trays placed outside every single nursery pod threshold stop boot-borne cross-contamination.

The LoRaWAN Sensor Nervous System

IoT Sensor Node	Target Boundary Threshold	Automated System Action Triggered	Target Outcome Saved
Ambient Room Temp	Drops below 30°C in Pods	Modulates biogas steam solenoids open	Piglet digestability & vitality
Ammonia (NH3) Gas	Exceeds 20 Parts Per Million	Triggers motorized curtain winch open	Respiratory health safety
Inline Water Flow	Drops 10% against baseline	Generates diagnostic dashboard alert	Early detection of subclinical illness

Infrastructure CAPEX Allocation Ledger

Asset Component Entry	Estimated CAPEX (ZAR)	Primary Engineering Objective
384 m^2 Poultry Sprint Barn Structure	ZAR 320,000	Galvanized frame with 50mm PIR roof insulation
4-Pod Swine Nursery Block (Polypropylene Slat)	ZAR 195,000	Physical 4-room biological isolation cells
30 kW / 30 kWh Off-Grid BESS & Solar Microgrid	ZAR 420,000	Electrical sovereignty & operational continuity
Biogas Digester & Engine Co-Generation Loop	ZAR 185,000	Replaces BESS heating draw with sub-floor steam heat
Solar Borehole & Elevated Tower	ZAR 95,000	5-Day safety hydration and slurry buffer

Baseline Financial Projections

Conservative Tariff Matrix

To establish clean financial models for Gustav, Senzo, and community investment partners, we run off-grid calculations:

- **Baseline Power Tariff:** Model assumes ZAR 1.80/kWh across all systems.
- **Internal Replacement Fund:** 100% of tariff savings are placed into capital replacement reserves.
- **Regional Alignment:** Collaboration with Precision Alliance and Mam Dube NPOs.

Social Imperatives

Operational blueprints integrate accredited green economy skills transfer pathways for local youth and women in Kinross and Secunda.

STEWARDSHIP LOCKED

Ready for Deployment. Engineered for Yield.

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Kinross & Secunda Regional Green Innovation Core